

PACKAGE HVAC UNIT INSPECTION REPORT

Client-Facing Condition Assessment

Equipment Type	Packaged HVAC Unit
Compressor	Copeland Scroll
Approximate Age	23 years
Operating Status	Operational at time of inspection

Scope / Components Reviewed

- Refrigeration system and accessible refrigerant piping
- Condensate drain lines and P-trap
- Air filters
- Heating section / heat exchanger
- Condenser coil
- Cabinet, equipment base, and accessible support structure
- Accessible mechanical and electrical components

Inspection Findings

1. Refrigeration System - Further Evaluation Required

Symptoms consistent with a potentially low refrigerant condition were observed during testing. Low refrigerant charge can reduce system capacity and efficiency and may indicate leakage within the refrigeration circuit. Refrigerant charge should not be adjusted without first evaluating the system for leakage and confirming operating conditions.

2. Heat Exchanger / Heating Section - Corrosion Observed

Visible corrosion and deterioration were observed within the heating section/heat exchanger area. Further evaluation by a qualified HVAC contractor is recommended to determine the extent of corrosion and whether repair or replacement is required.

3. Condensate Drainage / P-Trap

The condensate drain/P-trap configuration was found reversed or improperly oriented at the time of inspection. The configuration was corrected to establish proper drainage fall. The remaining condensate discharge piping should be evaluated and extended to discharge water away from the equipment and supporting structure.

4. Ponding / Drainage Conditions

Standing water and evidence of condensate accumulation were observed beneath and around the unit. Continued ponding can contribute to corrosion of the equipment base, support frame and surrounding materials and may contribute to deterioration or settlement over time.

5. Equipment and Support Corrosion

Visible corrosion is present at portions of the unit and supporting steel structure. Continued exposure to moisture may accelerate deterioration. Affected areas should be evaluated, cleaned and protected as appropriate.

6. Air Filtration / Maintenance Condition

The filters showed substantial accumulation of dirt and debris. Restricted filters can reduce airflow, decrease cooling performance, increase energy consumption and place additional load on blower and refrigeration components. Filter replacement and routine preventive maintenance are recommended.

7. Condenser Coil Condition

The condenser coil exhibited visible dirt/debris accumulation and general deterioration consistent with extended service. Coil cleaning and inspection are recommended to maintain proper heat transfer and condenser airflow.

Recommended Corrective Services

- Perform a complete refrigerant leak inspection of accessible refrigerant piping, coils, fittings and associated components.
- Verify refrigerant charge using appropriate operating measurements and manufacturer specifications.
- Extend/reconfigure the condensate drain discharge to direct water away from the equipment base and eliminate recurring ponding.
- Confirm proper condensate trap configuration, slope and drainage after correction.
- Replace or service dirty air filters and establish an appropriate replacement schedule.
- Clean and evaluate the condenser coil.
- Evaluate visible corrosion of the heat exchanger/heating section.
- Inspect the equipment cabinet, drain pan, base and supporting steel for deterioration associated with prolonged moisture exposure.
- Perform a comprehensive HVAC operational assessment to establish the condition of major components and determine whether continued repair is economically practical.

Equipment Age & Replacement Considerations

Although the packaged unit was operational at the time of inspection, operational status alone should not be interpreted as an indication of remaining service life or future reliability. At approximately 23 years of age, the equipment is in an advanced stage of its service life, and replacement planning is appropriate.

Advanced Equipment Age: Approximately 23 years of service increases exposure to age-related component deterioration and unplanned failure.

Energy Efficiency: Older equipment may operate at lower efficiency than modern replacement equipment, resulting in higher electrical consumption and operating costs.

Refrigerant Considerations: The refrigerant type should be confirmed from the equipment data plate. If the system utilizes R-22, refrigerant availability, cost and serviceability should be considered when evaluating future repairs.

Corrosion and Deterioration: Visible corrosion was observed in portions of the equipment and support structure. Continued deterioration can affect reliability and eventually require significant repair.

Aging Components: Motors, controls, contactors, capacitors, compressor components and other electrical/mechanical parts are subject to age-related wear and failure.

Increasing Repair Requirements: As equipment ages, repair frequency and cost may increase, making continued investment less economical.

Parts Availability: Replacement parts for older equipment may become discontinued, more expensive or difficult to obtain.

Reduced Performance: Aging equipment may experience declining cooling capacity, airflow, humidity control or difficulty maintaining intended design conditions.

Insurance / Property Considerations: Equipment age and documented condition may be considered during property-condition assessments, maintenance planning or insurance underwriting. Specific requirements should be confirmed with the applicable carrier.

Overall Assessment

The packaged HVAC unit was operational at the time of inspection; however, multiple age- and condition-related deficiencies were observed, including indications warranting evaluation of the refrigerant system, corrosion within the heating section, deteriorated/dirty filtration, condensate drainage deficiencies, ponding around the equipment, and corrosion of the unit/supporting structure.

Considering the unit's approximately 23-year service age, existing deterioration, current maintenance needs and potential for additional age-related component failures, obtaining a replacement estimate in conjunction with the recommended diagnostic repairs is advisable. This will allow the client to compare continued repair and maintenance costs against replacement with modern, higher-efficiency equipment.

Priority Recommendation: Address the condensate drainage/ponding condition and evaluate the heat exchanger/heating section and refrigeration system. Due to the age and overall condition of the equipment, replacement should be included in near-term capital planning.

PHOTOGRAPHIC DOCUMENTATION



Photo 1 - Overall view of packaged HVAC unit and service setup.



Photo 2 - Condenser coil and equipment support frame; visible weathering/corrosion noted.

PHOTOGRAPHIC DOCUMENTATION



Photo 3 - Heating section/heat exchanger area showing visible corrosion.



Photo 4 - Rear/side equipment view showing condensate drain arrangement and support frame.

PHOTOGRAPHIC DOCUMENTATION



Photo 5 - Air filter condition at time of inspection; significant dirt/debris accumulation visible.



Photo 6 - Condensate P-trap/drain piping and ponding beneath equipment.